

CASE STUDY IN AI FOR BUSINESS · MBA ·

AI-Powered Talent Intelligence & Internal Mobility at GB Corp

A skills-based framework for reducing external hiring dependency in a multi-business-unit group — with a working simulation

PRESENTED BY

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THE BUSINESS PROBLEM IN ONE SENTENCE

**We hire from outside for jobs
our own people could already
do.**

GB Corp · six lines of business · Egypt, Jordan and Iraq · about 9,300 employees · scope of this study: 3,200 white-collar professionals

THE EVIDENCE

The talent is already inside GB Corp

10%

of open roles are **posted internally**

48%

already have an internal candidate who is **ready now** — holding 80%+ of the required skills

75%

of those ready-now people sit in a **different business unit** from the role

12.3%

of **high performers** left last year — the most expensive attrition

61%

external hires are more likely to be let go than internal promotes (Bidwell, 2011)

ROOT CAUSE

The HR system records positions, not capabilities

WHAT THE HR SYSTEM CAN SEARCH

Job title	Grade
Cost centre	Reporting line
Location	Hire date

Structured, indexed — and says nothing about what a person *can do*.

WHAT PREDICTS ROLE FIT — TRAPPED IN PROSE

CV excerpt · Senior Officer, GB Capital · G6

“Six years in consumer finance. Built the monthly provisioning model in Excel and rebuilt the collections dashboard. Handled IFRS 9 reporting for the leasing book. Trained two junior analysts. Arabic and English.”

So the question “who here could already do this role?” is never asked.

WHAT THIS PROJECT DELIVERS

One question, four contributions

Research question: can AI make internal capability visible enough to shift hiring inside the group — explainably, fairly, and at a return that justifies the cost?

1

A four-layer architecture

Data → semantic → intelligence → action. Each layer is useless without the one beneath it.

2

A five-level maturity ladder

Positions GB Corp today and defines the next step. Rungs cannot be skipped.

3

A locally calibrated valuation

Every cost expressed as a share of our own salary base — not imported US benchmarks.

4

A working simulation

Built in Google AI Studio. Every step from CV to ROI can be inspected and challenged.

CASE & DATASET

A multi-unit group — where the problem is sharpest

THE CASE ORGANISATION

Listed Egyptian automotive and mobility group. Six lines of business:

- Passenger cars
- 2-, 3- and 4-wheelers
- Commercial vehicles & equipment
- After-sales & service
- Trading & spare parts
- Financial services (GB Capital)

THE DATASET — FIVE LINKED TABLES, ~4,000 ROWS

Table	Rows	What it holds	Label
Employees	3,200	Grade, tenure, promotion history, engagement, compa-ratio, skills	Attrition
Open roles	90	Live requisitions across all six divisions, with required skills	—
Candidate matches	450	Top five internal candidates per role	Readiness
Skills master	66	Supply against demand per skill — 46 in shortage	Gap signal
Internal moves	240	Past moves and whether the mover stayed	Still employed

Four layers, each built on the one below

4 · ACTION

Ranked shortlist to the hiring manager, matched opportunity to the employee — inside the HRIS. A human decides.

3 · INTELLIGENCE

Attrition classification (supervised) + skill-similarity matching (unsupervised) → readiness band and named gap.

2 · SEMANTIC

NLP extracts skills from CVs, JDs and training records; maps them to one controlled taxonomy.

1 · DATA

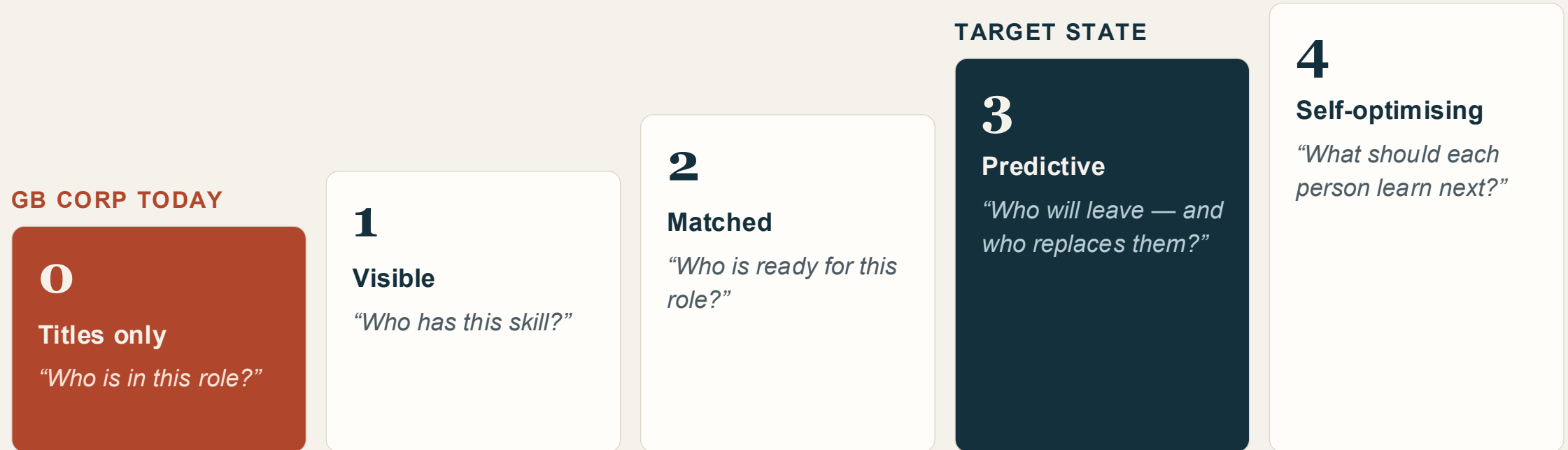
Employee-owned Talent Marketplace profiles, HRIS master data, open roles, move history.

THE DESIGN DECISION THAT MATTERS MOST

The output is a **named gap**, not a **score**.

A score can only be accepted or rejected. A named gap is a development plan a line manager can approve the same day — and it makes the recommendation contestable.

The Talent Intelligence maturity ladder



You cannot skip a rung. Matching is undefined without profiles; prediction needs the movement history that matching creates. **So we start with the marketplace, not the AI.**

Four modules, one dataset

An interactive application built in Google AI Studio on the project dataset. It was built so that the path from a CV to a financial conclusion can be inspected at every step — and every assumption can be changed live.

1

How the AI works

One CV traced through six auditable stages

2

Live matching

3,200 employees scored against 90 open roles

3

Flight risk

The attrition model, with its formula on screen

4

ROI simulator

Eleven assumptions on sliders, recalculated live

MODULE 1 · HOW THE AI WORKS

Normalisation turns documents into data

TAB 1 — HOW THE AI WORKS

TAB 2 — LIVE MATCHING (90 ROLES)

TAB 3 — FLIGHT RISK & WATCHLIST

TAB 4 — ROI SIMULATOR

How the AI Works — End-to-End Pipeline

Auditable transformation of unstructured employee resumes into canonical skill vectors, transparent scoring, and managerial decisions.

1 Raw Input

2 Extraction

3 Normalisation

4 Vectorise

5 Score

6 Decide

STEP 3 OF 6

Taxonomy Normalisation & Semantic Grounding

Standardisation

Key Architectural Concept: Without this step, "Excel modelling" and "financial modelling" are two unrelated strings and no matching is possible.

Raw Extracted Phrase	Mapping	Canonical Skill (Taxonomy)	Skill Category
"provisioning model in Excel"	→	Financial Modelling	Financial & Risk
"collections dashboard"	→	Collections Management	Financial & Risk
"IFRS 9 reporting"	→	IFRS Reporting	Financial & Risk
"trained two junior analysts"	→	Learning & Development	Leadership & Org
"Arabic and English"	→	Arabic-English Business Communication	Leadership & Org

- 1 · Raw input
- 2 · Extraction
- 3 · Normalisation
- 4 · Vectorise
- 5 · Score
- 6 · Decide

Without this step, “Excel modelling” and “financial modelling” are two unrelated strings — **and no matching is possible.**

MODULE 1 · TRANSPARENT SCORING

Every score is arithmetic a manager can check

TAB 1 — HOW THE AI WORKS

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How the AI Works — End-to-End Pipeline

Auditable transformation of unstructured employee resumes into canonical skill vectors, transparent scoring, and managerial decisions.

← Previous Step

Next Step →

Replay

1 Raw Input

2 Extraction

3 Normalisation

4 Vectorise

5 Score

6 Decide

STEP 5 OF 6

Transparent Deterministic Scoring (No Black-Box)

Formula Auditability

The matching score is calculated via transparent arithmetic combining skill coverage, performance rating bonus, job family alignment, and grade difference:

Base Skill Coverage: 5 of 7 skills matched → coverage 71% (71 pts)

Performance Rating Bonus: Rating 4/5 (Exceeds expectations) → +6 pts

Job Family Synergy: Same domain (Finance & Risk) → +4 pts

Grade Seniority Delta: 1 grade below target (G6 vs. G7 target) → -5 pts

= Final MatchScore: 76 / 100

READY WITH DEVELOPMENT

Identified Skill Gaps for Targeted L&D: Portfolio Risk Analysis · AML / KYC Screening

Component	Basis	Pts
Base skill coverage	5 of 7 skills	71
Performance bonus	rated 4 of 5	+6
Job-family synergy	same domain	+4
Grade delta	G6 vs G7	-5
Match score	Ready w/ dev.	76

NAMED GAP
Portfolio Risk Analysis · AML / KYC Screening

MODULE 1 · THE DECISION

One computation, two conversations

TAB 1 — HOW THE AI WORKS

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How the AI Works — End-to-End Pipeline

Auditable transformation of unstructured employee resumes into canonical skill vectors, transparent scoring, and managerial decisions.

← Previous Step

Completed (Replay)

Replay

1 Raw Input

2 Extraction

3 Normalisation

4 Vectorise

5 Score

6 Decide

STEP 6 OF 6

Human-in-the-Loop Dual Decision Outputs

Operationalization

TO THE HIRING MANAGER

Ranked Shortlist

Requisition REQ-001: Credit Risk Analyst

5 qualified internal candidates discovered across 3 business units:

#1 Ahmed Hassan (GB Capital • G6)76 ptsReady w/ Dev

#2 Karim Mostafa (GB Auto Passenger • G7)88 ptsReady Now

#3 Sara El-Sayed (Tires & Aftersales • G6)72 ptsReady w/ Dev

Hiring manager retains full autonomy to invite for interview, reject, or request external candidates.

TO THE EMPLOYEE (AHMED HASSAN)

Career Pathway

Career Opportunity Match

Personalized career trajectory recommendation:

"You are 71% ready for Credit Risk Analyst — two skills away."

Your background in Financial Modelling, Collections, and IFRS 9 makes you a strong contender for promotion to G7.

Suggested Learning Path to Close the 29% Gap:

- Portfolio Risk Analysis: GB Academy 3-week simulation
- AML / KYC Screening: Central Bank of Egypt compliance module

✓ Notification pushed to employee self-service portal

The AI recommends.
A person decides.

To the manager: five qualified internal candidates across three business units.

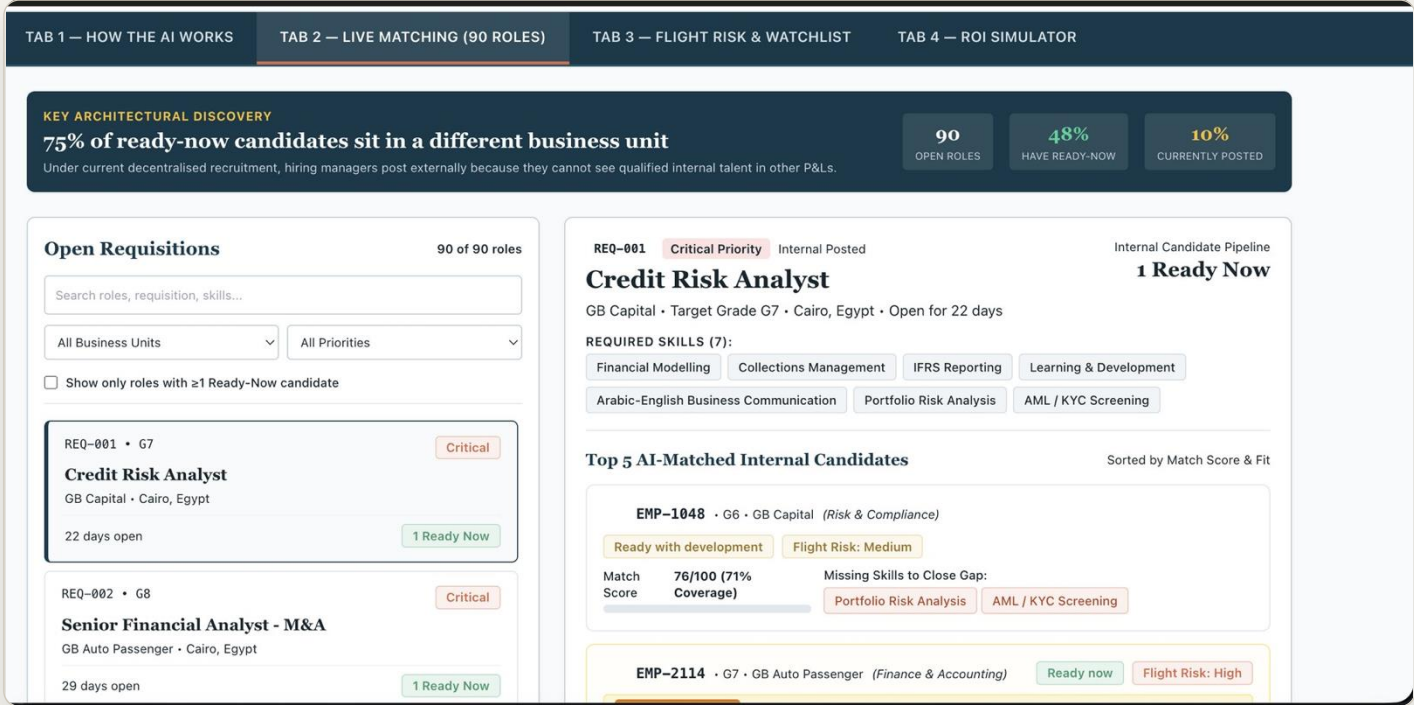
To the employee: “You are 71% ready — two skills away,” with a learning path for each gap.

The best candidate — **88 points, ready now** — sits in a different business unit.

12

MODULE 2 · LIVE MATCHING

Across 90 roles, the qualified people are next door



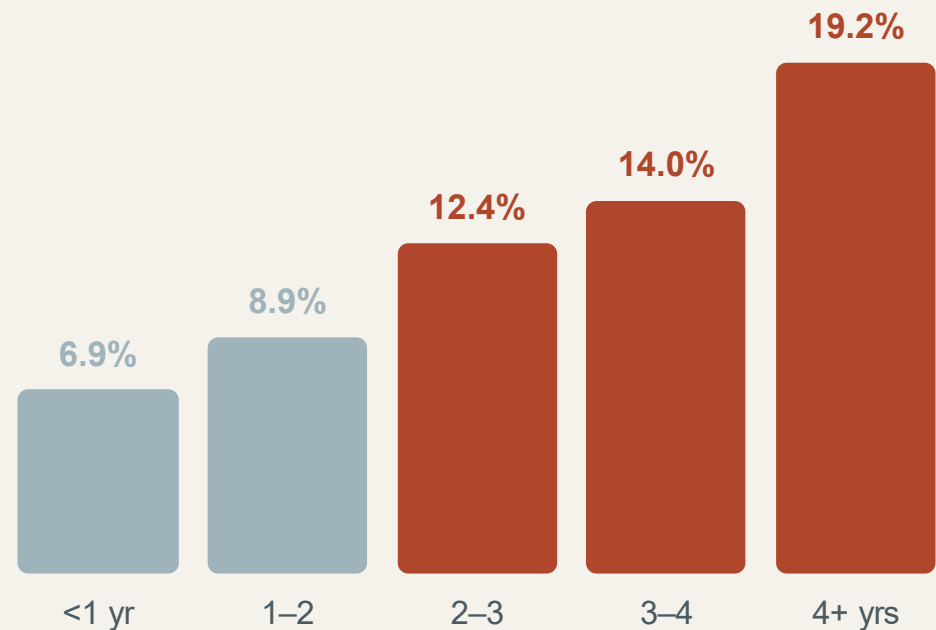
75%

of ready-now candidates sit in a **different business unit** from the role.

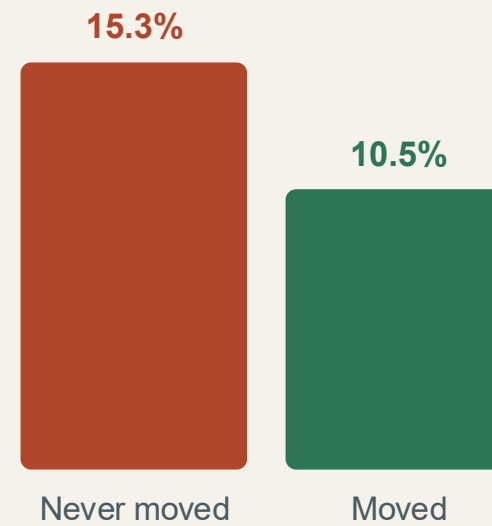
Managers are not choosing outsiders over insiders. **They cannot see the insiders.**

Stagnation predicts resignation; movement prevents it

Attrition by years since last promotion



Internal move history (tenure 2+ yrs)



WHERE THEY MEET

2.3×

Stagnating high performers resign at **25.4%** vs 11.2% for everyone else.

122 people — under 4% of scope, identifiable in advance.

What GB Corp would invest

YEAR 1 — BUILD YEAR (EGP)

Item	EGP
Talent Marketplace licence (3,200 users)	1,200,000
HRIS / ERP integration	900,000
AI model + skills taxonomy	700,000
Change management, training, comms	550,000
Data migration — profile pre-population	450,000
Project management	200,000
Total year 1	4,000,000

YEARS 2 AND 3 — RUN COST, EACH (EGP)

Item	EGP
Platform licence	900,000
Model maintenance / retraining	180,000
Administration and support	120,000
Per year	1,200,000

TOTAL 3-YEAR INVESTMENT

EGP 6.4M

4.0M + 1.2M × 2

Where the money comes back

3,200

white-collar staff

×

11.8%

attrition

=

378

vacancies / yr

×

+15 pts

fill rate 10→25%

=

57

extra internal fills

VALUE OF ONE SWITCHED HIRE (EGP)

Recruitment cost avoided	20% vs 4% of salary	69,040
Salary premium avoided	18% external premium	77,670
Vacancy time saved	35 days × 50% productivity	29,040
Retention dividend	4.8 pp lower attrition	4,140
Per switched hire		≈179,900

57 hires × EGP 179,900 =

EGP 10.2M

benefit per year at full run. Year 1 counted at 60% (6.1M) while adoption ramps.

EGP 3.14 back for every pound invested

EGP 20.1M

three-year net value

314%

return on investment

7.9 mo

payback — inside year 1

EGP	Year 1 (60%)	Year 2	Year 3	Total
Benefit	6,113,546	10,189,243	10,189,243	26,492,033
Investment	(4,000,000)	(1,200,000)	(1,200,000)	(6,400,000)
Net	2,113,546	8,989,243	8,989,243	20,092,033
Cumulative	2,113,546	11,102,790	20,092,033	

ROI = 20,092,033 ÷ 6,400,000. Payback = 4,000,000 ÷ (6,113,546 ÷ 12). Modelled, not measured — every figure is a live formula.

One decision, three outcomes

FAILURE CASE

**Fund it, change
nothing**

Internal fill rate: 10%

−6.4M

3-year net value

ROI **−100%** · payback **never**

0 extra internal moves / yr

RECOMMENDED

Target

Internal fill rate: 25%

20.1M

3-year net value

ROI **314%** · payback **7.9 mo**

57 extra internal moves / yr

STRETCH

Best-in-class

Internal fill rate: 40%

46.6M

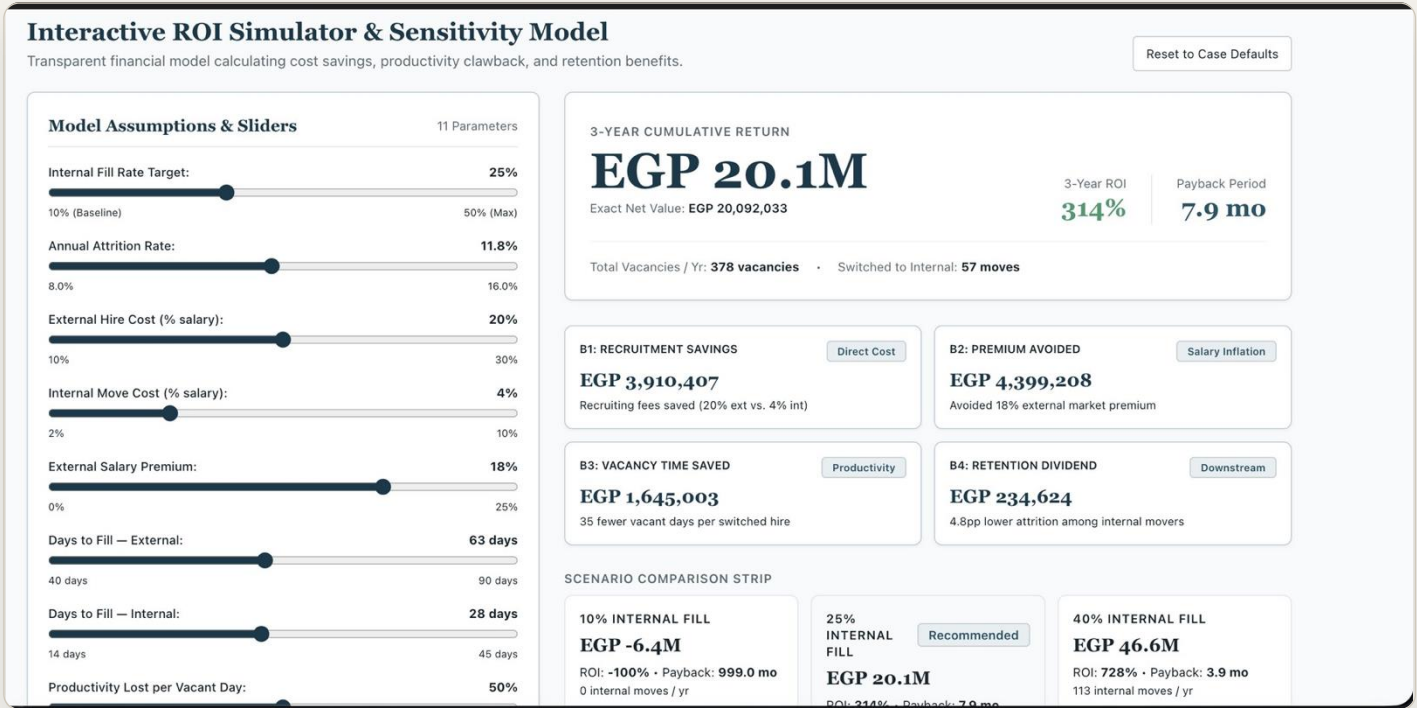
3-year net value

ROI **728%** · payback **3.9 mo**

113 extra internal moves / yr

Only the fill rate changes; everything else is held constant. **25% sits inside the documented top-quartile band** — defensible from a Level 0 start. And the left column shows **the return depends on adoption, not installation.**

LIVE DEMONSTRATION



Compliance is built into the design

Obligation	Where the design answers it
Human oversight — no solely automated decision	System recommends only; manager can interview, reject or go external
Right to explanation	Match score shown as four named components; driver shown per person
Transparency to the person affected	Employee sees the same result as a career path with named gaps
Consent in profiling	Profiles are written and owned by the employee
Bias testing before deployment	Not yet done — required before any live use

IMPLEMENTATION

Two conditions will decide success

1

Manager incentives

If losing a strong team member to another division feels like a personal cost, managers will block moves — whatever the system recommends.

Fix: count talent released to other units in the releasing manager's own performance scorecard.

2

Profile completion

A marketplace filled by a minority matches a minority of roles. Failure at Level 1 closes every level above it.

Fix: reciprocity — complete your profile to see opportunities — and pre-populate profiles so people edit, not write.

Neither risk is technical. **Both decide the outcome more than any modelling choice.**

MEASUREMENT

How we will know it worked

Business KPI	Baseline	Target	Change
Internal fill rate	10.0%	25%	+15 pts
Roles with a ready-now internal match	47.8%	60%	+12 pts
Regretted attrition (high performers)	12.3%	8%	-35%
Average requisition age	63 days	28 days	-56%
Stagnation rate (3+ yrs, never moved)	10.0%	6%	-40%
Critical roles with a ready successor	53.3%	65%	+12 pts

MODEL KPI THAT GOVERNS

**Recall on
leavers — not
accuracy**

88% of people stay, so a model
that predicts “nobody leaves” is
88% accurate — and useless.

Every baseline is a live formula in the project workbook — so the same formulas can re-measure the result.

CONCLUSION · THE CLAIM THIS PROJECT DEFENDS

Not the size of the return — but that the path from an unstructured CV to a financial conclusion can be traced, at every step, **by someone who did not build it.**

48% vs 10%

roles with a ready internal candidate vs roles posted internally

6.9% → 19.2%

attrition as time since promotion grows

EGP 20.1M

3-year net value · 314% ROI · 7.9-month payback

Thank you.

Questions and discussion

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BACKUP · VERIFICATION

Re-deriving the headline from four constants

Step	Derivation (EGP)	Result
Annual vacancies	$3,200 \times 11.8\%$	$377.6 \approx 378$
Switched to internal	$377.6 \times (25\% - 10\%)$	$56.6 \approx 57$
B1 Recruitment savings	$56.6 \times 431,498 \times (20\% - 4\%)$	3,910,407
B2 Salary premium avoided	$56.6 \times 431,498 \times 18\%$	4,399,208
B3 Vacancy time saved	$56.6 \times 35 \text{ days} \times 50\% \times (431,498 \div 260)$	1,645,003
B4 Retention dividend	$56.6 \times 4.8\text{pp} \times 431,498 \times 20\%$	234,624
Total annual benefit	$B1 + B2 + B3 + B4$	10,189,243
3-year benefit	$60\% \times Y1 + Y2 + Y3$	26,492,033
3-year net value	$26,492,033 - 6,400,000$	20,092,033
ROI · payback	$\div 6.4\text{M} \cdot 4.0\text{M} \div (6.11\text{M} \div 12)$	$314\% \cdot 7.9 \text{ mo}$

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